

Achieving supply chain excellence through supplier management

A case study of fast moving consumer goods

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Abstract

Purpose – The purpose of this paper is to develop a supplier selection and management program to improve overall supplier performance.

Design/methodology/approach – Supplier performance is measured in terms of quality and delivery within a fast moving consumer goods (FMCG) business of a multinational company based in Thailand using a case study methodology. The quality and delivery related data were collected from daily deliveries at the manufacturing plant both before and after implementing the supplier management program.

Findings – Findings of the study suggest that the selection of suppliers based on their performance is important for manufacturing firms. Moreover, the supplier selection and management program can contribute effectively to improving suppliers' performance.

Research limitations/implications – This case study has been conducted based on a single company within the FMCG industry. Hence, it limits the generalizability of the findings across industries.

Practical implications – The study provides a real-life tool for practitioners to learn about the importance of strategic decision-making process pertaining to the supplier selection and management program.

Social implications – This study demonstrates that through a transparent supplier evaluation process, the firms can develop trust and long-term relationship with their suppliers for pursuing the goals of product development and innovation.

Originality/value – Implementing a supplier management system is a critical step in enhancing an organization's overall competitiveness. To develop an effective supplier management system firms must have objective measures and share those with their suppliers. Developing metrics for suppliers' evaluation is the key to achieving continuous improvement as evidenced in this case.

Keywords Thailand, Qualitative research, Supply chain management, Case study, Supplier selection, Supplier management, Fast moving consumer goods

Paper type Case study

Introduction

Organizations have discovered that the efficiency of their supply chain determines their competitiveness in the marketplace (Sjoerdsma and van Weele, 2015), and the firms' ability to manage the supply chain resources and its suppliers determines their capacity to deliver the right products at a faster pace and a cost lower than their competitors (Lii and Kuo, 2016). In other words, the supply chain excellence is linked with its suppliers' base. The supply chain performance directly depends upon the performance of its suppliers (Forslund, 2014), and in the Fast Moving Consumer Goods (FMCG) business, the substitution of products is always happening at a fast pace, and the role of supplier management in the new product development is needed to compete effectively. Supply chain excellence is a relentless effort in achieving a context-specific efficiency based on a strategic framework driven by a set of underlying principles (Lapide, 2006). Supplier management is one of the key areas that supply chains emphasize because the materials supplied is the entry point of the chain (Hsu *et al.*, 2016), and the poor performance in either delivery or quality of supply leads to a poor response to the customer requirement.

Evaluating and selecting suppliers emerges as a central issue in executing the supplier management function while developing a right purchasing strategy (Kumar and Routroy, 2017). Supplier evaluation and supplier management decisions have acquired an increased



importance and a sound selection decision reduces or prevents a host of problems. Competitive bidding used to be a prominent method for awarding purchase contracts in the past but now, multinational firms in the FMCG industry are committing their major resources to evaluate the suppliers' performance and their capability across many areas (Forslund, 2014). The supplier selection process has become critical and teams of cross-functional personnel in organizations being seen visiting and evaluating suppliers these days is a commonplace (Morita *et al.*, 2015). In this research, the selection and management of suppliers are studied which is critical for the organization to survive and excel. The underlying research questions of this study are to address:

RQ1. Does supplier selection and performance measurement help improve suppliers' performance?

RQ2. How to develop suppliers' selection and management system?

The following part of the paper presents a detailed review literature and throws light on addressing the issues of how to develop an effective supplier selection and management system.

Literature review

Supplier selection and management: an introduction

Supplier selection and their management has become a centric proposition for manufacturers to maintain resilience against disruptions (Chopra and Sodhi, 2014), and companies now recognize that supplier management plays key role in achieving long-term competitive advantage of the firm. It has become an important tool for manufacturing companies and it contributes significantly to the achievement of manufacturers' performance (Poppo *et al.*, 2016). An effective supplier management delineates with the organization of the optimal flow of high quality, value-for-money materials or components to manufacturing companies from a suitable set of innovative suppliers (Pulles *et al.*, 2016). Close co-operation with the suppliers helps in bringing down unit cost (Kırlmaz and Erol, 2017) and ensuring the highest quality of supply at the lowest cost in the long term (Stevens and Johnson, 2016). Good suppliers also help manufacturers in the development phase which results in cost reductions and enhanced delivery performance (Lee and Schmidt, 2017). Morita *et al.* (2015) argued that the supplier selection and management has an important role and there are two key issues to be considered: the trend in the supplier base – the tendency for leading companies to reduce their number of suppliers (due to the advantages it offers), and manufacturers' sourcing policies – their willingness to reduce their supplier base to the extent that they mainly have "single-source" suppliers. Current management thinking also recommends that the establishment of long-term relationships with a reduced number of suppliers play a critical role in contributing to the new product development by reducing costs and continuously improving quality (Uluskan *et al.*, 2016). On the other hand, evaluation and management of suppliers is the process of measuring, analyzing and managing supplier performance to improve quality, reduce cost, mitigate supply risk and drive continuous improvement in the supplied value (Forslund, 2014; Nguyen *et al.*, 2017).

To develop a resilient supply chain and to cope up with the uncertainty in the sourcing strategies, the important factors considered are single and multiple sourcing, backup suppliers' contracts, spot purchasing, collaboration and supply chain visibility (Namdar *et al.*, 2018). Single sourcing uses only one source of supply for a particular material or a component part (Jia *et al.*, 2017) and, it is claimed to lead to improved quality and better pricing strategies (Dabhilkar *et al.*, 2016). It also results in a closer relationship with supplier churning more tangible benefits such as fewer incoming defective parts, greater delivery

reliability and lower cost. In contrary, multiple sourcing allows for flexibility and alternative choices (Tsai, 2016). Stevens and Johnson (2016) are of the view that single source suppliers normally delivered higher quality at lower cost to the buyers and recommend that companies can benefit from a single sourcing approach. However, Schmitt and Singh (2012) find risks involved in it and propose that an emergency plan should always be in place to deal with the challenges of the single source suppliers. If a firm reduces its supply base considerably and its most suppliers receive longer-term agreements, the willingness or ability to switch suppliers is diminished in the firm's value chain. Thus, achieving an effective supplier management is a critical area for manufacturers, and likely to remain important in the time to come. Uluskan *et al.* (2016) argue that suppliers' performance measurement is an important tool and manufacturers must invest considerable management resources in improving or standardizing its efficacy. Academic literature offers only a few empirically consolidated findings that provide guidance on the kind of measures to choose and the form of the supplier selection and management process that should be implemented in the inter-organizational context (Spekman and Davis, 2016).

Supplier selection and management strategies

Supplier selection strategy is the strategy adopted by the manufacturer to evaluate and select suppliers which fulfills the requirements of the manufacturer (Mafakheri *et al.*, 2011), and quality, cost and delivery are the three most important criteria in the suppliers' selection (de Boer and de Boer, 2017). Selection of suppliers is critical and the trend toward "Just-in-Time" manufacturing practices has resulted in a supply base reduction (Prajogo *et al.*, 2016). There is a greater need for interaction between the buyers and suppliers due to resource scarcity, and firms involve their suppliers early in the process to deliver superior value to their customers (Peng and Lu, 2017). To release products quickly, supplier selection occurs at the front end of the program, long before the specification is laid out (Guinot *et al.*, 2016). Leading companies involve suppliers by reviewing their technology roadmaps to determine what technology is emerging that they might be able to use in the future products (Tajeddini, 2016). By bringing the market-leading technologies into the design process at an early stage, suppliers can help the manufacturer in reducing the lead time for the improvement of the product's performance. On the other hand, Ye and Lau (2018) found suppliers can better assist manufacturers in optimizing the adoption of new technologies to improve their products' launch and in response to demand fluctuations. The supplier selection strategy regarding technology, quality, cost and delivery performance are important strategies for overcoming the "upstream uncertainties" such as suppliers' lapse on the delivery and performance, high-cost production, and quality rejections (Oghazi *et al.*, 2016). It also helps in dealing with the challenges of "downstream uncertainties" arising due to the demand volatility and changes in the product mix, price and competitive actions which requires flexibility in the manufacturing processes (Arora *et al.*, 2016).

Supplier management strategy is pursued by the manufacturer to improve its supplier's performance and capabilities to meet the manufacturer's short-term and long-term supply needs (D'Amico *et al.*, 2017). Supplier management is concerned with organizing the optimal flow of high quality, value-for-money materials or components to a manufacturing company from a suitable set of innovative suppliers (Pulles *et al.*, 2016). This strategy allows the manufacturer to bridge the gap between its suppliers' capabilities and their expectations from the suppliers (Wilhelm *et al.*, 2016). It is impossible to find the supplier who matches 100 percent of the manufacturer's expectations in the initial selection process. Involving suppliers in the product development activities and in the continuous improvement initiatives help them learn more about customers' requirement and to become more efficient in meeting the manufacturers' expectations (Manders *et al.*, 2016). These strategies help organizations enhance communication with the suppliers, share knowledge, improve

decision-making, and upgrade suppliers' performance. Prajogo *et al.* (2016) find that although both the supplier selection criteria and the supplier involvement are positively correlated with manufacturing performance, the supplier involvement in product design activities and continuous improvement efforts is much lower than the use of supplier selection criteria. Supplier involvement has an even greater benefit if it is undertaken earlier in the process, and helps to shorten the design cycle time which means faster launch flexibility for the firm (Godinho *et al.*, 2017).

Suppliers-manufacturers relationship

The relationship between suppliers and manufacturers is an important one (Morita *et al.*, 2015), and suppliers have a substantial and direct impact on the cost, quality, technology, and time-to-market of the new products (Wiengarten and Ambrose, 2017; Zaremba *et al.*, 2017). The management of supplier relationship is a vital task for manufacturers and it contributes to both the competitiveness and profitability of the company (Pradabwong *et al.*, 2017). As time-to-market continues to get compressed due to customers' changing demands, manufacturers respond with a complementary reduction in the product lifecycles (Hsu *et al.*, 2016). To meet this variability of demand, the manufacturer needs to incorporate flexible supply chains into their operations that are scalable and adaptable to achieve shorter order fulfillment lead time. Only firms with advanced technology having a competitive edge can overcome stiff competition by introducing a wide range of products to meet the different market segments and be able to deliver quickly into the hands of customers before any of its competitors can do so (Bruque-Cámara *et al.*, 2016). By selecting a supplier with leading-edge technology, the manufacturer can leverage on its competency to introduce more products and enjoy the first movers' advantages. The selection of suppliers based on quality is more important compared to technology, cost and delivery performance. By involving suppliers in the early stage of design, having clear quality and technology roadmaps, and a proper inventory management program, the manufacturer gets better equipped to combat the war of uncertainty in its supply chain management. In many industries, the management of suppliers can account for a significant percentage of the total cost of manufacturing (Fredendall *et al.*, 2016). There is no denying to the fact that an effective supplier management system results in an optimal inventory level (Aitken *et al.*, 2016).

Supplier management system

An effective suppliers' evaluation and management system is critical for companies to be competitive in the global marketplace (Burki and Buvik, 2017) and research indicates that nearly 50 percent of the companies in different industries have a formal supplier management process (Huq *et al.*, 2016). The factors mostly used in the current supplier evaluation and management method are quality, supplier certification, facilities, continuous improvement, physical distribution and channel relationship (Hwang *et al.*, 2016). Su *et al.* (2016) observe that most supplier evaluation and management methods rely on the industry certification or heuristics indicators for evaluating the performance of suppliers. Suppliers' selection is no longer just the issue of obtaining low product prices. It deals with the costs in all related supply chain activities. An easy to use suppliers' evaluation model that covers SCM (supply chain management) issues can offer buyers great benefits in searching for new suppliers and evaluating the current suppliers.

Companies need to improve their supply chain operations continuously and add new suppliers to the existing supply chain as part of the improvement exercise. Even with the very competitive product prices, Wal-Mart is relentlessly searching for the suppliers who can provide products with lower prices and better services (Umashankar *et al.*, 2016). In the present context, selecting new partners and evaluating the current and existing

partners becomes critical in the management and implementation of the supply chain operations. In that background, the present research aims to provide a model that is comprehensive and flexible enough to resolve one of the major strategic decision issues to have an effective sourcing operation in an FMCG organization. The literature on supplier management shows that supplier-manufacturer relationship has evolved and a partnership approach is now preferable. Partnerships are characterized by closer contacts and are sometimes embedded in the manufacturers' single sourcing strategies. A long-term perspective is a prerequisite for a higher degree of integration among partners and a greater level of information sharing. Purchasing managers become information brokers as they provide the interface between the manufacturers and their supply base to perform suppliers' selection and evaluation to maximize the benefits of both parties. Use of a cross-functional team is needed to ensure that the selection and evaluation outcomes fit with the organizational strategies, and creates a balance in involving its stakeholders for putting in place an effective suppliers' management strategy. The involvement of the suppliers and engaging them is a key issue and organizations look forward in forging a long-term relationship with them.

Research methodology

In today's uncertain business environment, suppliers make a major contribution to a manufacturer's performance in the areas of quality, delivery, cost containment and new product development. Consequently, effective supplier management has evolved as a critical issue for the organizations to sustain their competitiveness by contributing towards an effective supply chain management. The research was carried out using a case study methodology in line with Yin's approach. This helps us to answer "how" and "why" of a process where there is little control over the events and the focus remains on the real-life situation (Yin, 2012, 2014). An experimental case study approach focuses on developing a suppliers' selection and suppliers' evaluation scheme that results in the performance improvement regarding delivery and quality (Salam and Khan, 2016). The present study provides a detailed account of the analysis of the supplier selection and management program undertaken by a multinational FMCG company based in Thailand. It is a subsidiary of one of the largest MNC consumer goods companies operating in 150 countries. This company has been selected due to the fact that, it is committed to the same spirit and vision of its parent company to satisfy the daily needs of the Thai people by meeting their needs with high-quality products and services. To avoid the original identity of the firm, a pseudonym, Mekong Thai Holding Company was used. The case study helps us to understand and analyze how an effective supplier selection and management system can lead to a better supplier performance. This case study was conducted in 2015 based on the company's original data and analyzed the supplier selection and management program in the light of the procurement and supply management principles. The research data were collected over a period of six years' time (years 2010–2015) from the locations where the materials arrive at the facility. It included the delivery data regarding the suppliers' responsiveness in adhering to the appropriate quality and quantity requirement. It also included the data about rejections and incomplete supply done by the supplier.

The data were collected by recording performance of each delivery. The delivery was checked for its scheduled arrival (on time-OT) and desired quality and condition (quality in full-QIF) of its contents *vis-à-vis* the purchasing order of the firm. For rejection, the materials were sampled and evaluated against the prescribed specification. Delivery not conforming to the product development specification was termed as the rejection. The firm uses a cross-functional team comprising of purchasing, quality assurance, logistics and R&D departments which evaluates the suppliers' selection and management.

Price, quality, delivery and technical support were chosen as the suppliers' selection criteria whereas price, quality, delivery, technical support and financial documentation were criterion considered for their evaluation. Key suppliers' selection criteria were developed based on the annual value spend, location, company's business strategy and its compatibility, and most importantly the suppliers' top management commitment to co-operate in the program. The evaluation criteria are communicated to the key suppliers while implementing it. The cross-functional team reviews the key suppliers' performance and decides about the suppliers' continuation with the firm. The performance of the key suppliers is analyzed by comparing the before and after status of the suppliers' performance under the evaluation program taking into account the On Time and Quality In Full (OTQIF) percentage.

Analysis and findings

As found in the case study, the supplier management program is an annual supplier assessment and management program at Mekong. The assessment is based on the input of the cross-functional team consisting of quality assurance, logistics, product development, accounts and purchasing departments. The results from the program identify suppliers' performance and provide opportunities for their improvement. This program helps in building long-lasting relationships with the suppliers based on the performance and excellence. The number of suppliers in the program is rationalized on a yearly basis based on their performance.

The supplier management program entails involving all stakeholders. It lists out the responsibility of the firm and suppliers both to achieve the required objectives of the program. These responsibilities are listed below:

Company's responsibility to the program

- (1) Encouraging the use and involvement of "preferred supplier" from the conceptual stage. At the Mekong, the key supplier gets the "preferred supplier" status from the assessment, and Mekong Thai Holding Company encourages and let suppliers get involved in all of its new projects from the conceptual stage in order to reduce the lead time to produce and market the new products.
- (2) Defining precisely the quality requirements and specifications to the suppliers.
- (3) Encouraging and promoting supplier's innovation and effort to improve quality.
- (4) Conducting the business in a fair and ethical way.
- (5) Ensuring that the program and its performance measurement is acceptable to all suppliers.

Suppliers' responsibility to the program

- (1) commitment to adhere to the requirement of the consistent quality and service;
- (2) promoting the continuous improvement efforts;
- (3) competitive pricing of the product;
- (4) working consistently toward zero defects;
- (5) complying with the agreed quality requirement and specification; and
- (6) providing an appropriate process of controls and testing.

Benefit to suppliers

- (1) a better understanding of the company’s requirements;
- (2) Mekong Thai Holding Company gives priority to its preferred supplier if there is a new raw materials requirement in the future; and
- (3) scores from this assessment program can be used to set future targets and improvement plan.

The Mekong adopts a five-step process to manage its’ suppliers selection and management process. Starting from the suppliers’ selection (both new and existing suppliers), it provides information about their management and evaluation. This part of the study is discussed below which provides detailed analysis and information about each step of the supplier’s selection and management process (Figure 1).

Step 1: developing suppliers’ selection criteria. At the Mekong, when a company has a new raw materials requirement, its supply management team develops the supplier’s selection form with details of material requirements, i.e., materials name, suppliers name, price/unit, pack size, packaging type, etc. Later, it is forwarded to the concerned departments (e.g. quality assurance, R&D, logistics and purchasing). On that request, each department responds with its score in their concerned area. Table I provides the supplier’s selection criteria being used at the Mekong Thai Holding Company. It provides the various score and its’ description while evaluating the suppliers to be adopted by various departments (purchasing, R&D, quality assurance and logistics) at the Mekong.

The scoring criteria adopted by various departments at the Mekong are very important to ascertain the total score of a supplier. Price, product development, quality assurance and logistics are chosen dimensions for evaluating the firm’s appropriateness. The purchasing

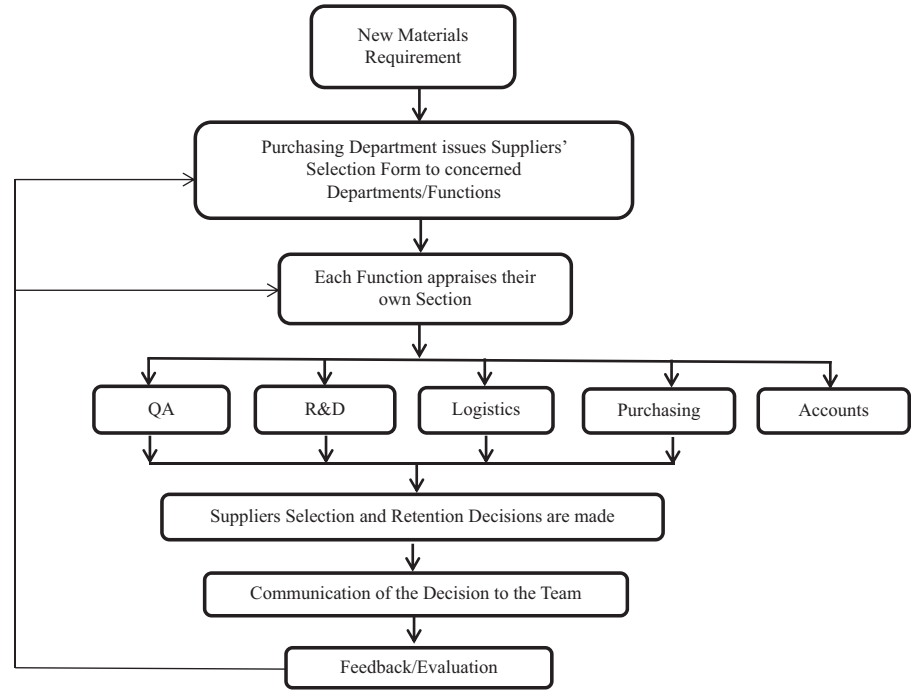


Figure 1.
Supplier selection
process overview at
Mekong Thai
Holding Company

Main criteria	Sub-criteria	Weights	Points	Scores and definitions
Purchasing	Price	15%	81–100	3 = Excellent (Exceeds the expectations)
	Pricing system	10%	61–80 41–60 40 or Below	2 = Good (Meets our expectations) 1 = Acceptable (Meets most of our expectations) 0 = Unacceptable (Needs improvement)
R&D	Product application	15%	81–100 61–80 41–60 40 or Below 81–100	3 = Excellent (Exceeds the expectations) 2 = Good (Meets our expectations) 1 = Acceptable (Meets most of our expectations) 0 = Unacceptable (Needs improvement)
	Technical support Certificate of analysis Process flow License (FDA) Material safety data sheet Technical know how Quality system (from audit)	10%	61–80 41–60 40 or Below	3 = Provided on time and complete document with practical and effective information 2 = Provided complete document on time 1 = Provided complete document but late 0 = Provided incomplete document and late
Quality assurance	Quality performance	15%	81–100 61–80 41–60 40 or Below	3 = Granted grade A 2 = Granted grade B1 1 = Granted grade B2 0 = Granted grade C
		10%	81–100 61–80 41–60 40 or Below	3 = No non-conformance/rejection 2 = 1–2 times of non-conformance or rejection 1 = 3–5 times of non-conformance or rejection 0 = > 5 times of non-conformance or rejection
Logistics	Delivery reliability	25%	81–100 61–80 41–60 40 or Below	3 = Excellent (Exceeds the expectations) 2 = Good (Meets our expectations) 1 = Acceptable (Meets most of our expectations) 0 = Unacceptable (Needs improvement)
	Lead time			

department considers the price (with due consideration to the fluctuation of the price, and price comparison with others in the same industry) and the pricing system (cost structure of the suppliers) as an important measure to grant the score to a supplier. Product development is an important dimension for the evaluation, and it includes the product application (comparison of product performance to the requirement, and its cost) and technical support (technical data which are required to be available and provided promptly by the suppliers as and when required). Quality assurance is another important criterion which contains quality system (suppliers' quality system data from audit result or pre-audit questionnaire) and quality performance (record of materials performance from particular suppliers) at the firm. Logistics also plays a very important role in choosing the suppliers at Mekong Thai Holding. Two factors are considered in logistics which are delivery reliability (timeliness and quantity of materials delivered to company warehouse against the purchase order) and lead time (as compared to others). After accumulating the scores obtained by the concerned departments, the qualified suppliers are shortlisted. The key suppliers are selected based on this list which is discussed in Step 2 given below.

Step 2: selecting key suppliers. While evaluating the selection criteria at Mekong Thai Holding, it has been found that the company uses four selection criteria to select its key suppliers which are as follows:

- (1) taking note of high annual spend (i.e. > 10m baht);
- (2) location of the manufacturing site in Thailand, traders were also included;
- (3) the potential of suppliers to develop and align with company's business strategy for the long run; and
- (4) suppliers' top management attitude and commitment level.

Step 3: developing key suppliers' performance measurement. The new supplier management program uses Key Performance Index (KPI) to measure its suppliers' performance. KPI measures on-time delivery with prescribed quality in full, i.e., OTQIF, and the complete order to be arrived on time with the quality satisfaction. The target to be achieved was 99.5 percent for the year 2015:

$$\begin{aligned} \text{KPI} &= \% \text{OTQIF (On Time Quality in Full)} \\ &= (1 - (\text{Missing Lot} / \text{Total received})) \times 100 \\ \text{Target} &= 99.5\%. \end{aligned}$$

A missing lot is divided into the delivery non-performance and rejection. Delivery performance is rated on its timely arrival as per the delivery plan and with no short shipping, i.e., the quantities of each delivery must be in line with the purchase order (variance up to 10 percent is allowed). Quality of the supply is very important at Mekong, and rejections are those orders which are not complying with the specification. The detailed description of % OTQIF used for the KPI is as follows:

$$\% \text{OTQIF} = (1 - (\text{No. of untimely supply} + \text{No. of short ship} + \text{No. of rejection}) / \text{Total lot received}) \times 100.$$

The percentage of the missing lot of the order is reviewed on a monthly basis to see the overall performance of the key suppliers, and it is shared with them to find ways for the improvement. The right first time or the OTQIF is calculated as mentioned above on a monthly basis. The times of missing lots are collected separately for all untimely, short

shipping and rejection orders. The graph illustrates the performance trend and achievement of the target which is kept less than 0.5 percent. Moreover, the company also ranks top suppliers on the dimension of missing lot which helps the company to focus on those key suppliers who still have problems, and helps them to improve in that regard.

From the performance monitoring, the company prescribes corrective and preventive actions for the suppliers as soon as the non-conformance occurs. The supplier must provide feedback on the valid action plan adopted in this regard. In the first quarter of the year 2015, there was one untimely delivery because of a tank car breakdown. The corrective action was taken immediately to fix it, and in the second quarter, there had been two incidents of short shipping because the documents were lost. The delivery had been immediately performed, and the preventive action was taken in this case by dedicating one person (by the supplier) to take care of the company's orders.

Step 4: supplier evaluation. Cross-functional team input is very important for evaluating and managing suppliers list. This team plays important role in arriving at the full score to be ascertained. There are five departments, i.e., quality assurance, R&D, logistics, accounts and purchasing who own the process and they evaluate them on eight chosen areas (see Table II).

Scoring criteria for each dimension:

- (1) Quality: it is divided into three areas namely: quality system, products' quality and its performance, and quality improvement and responsiveness of the suppliers:

- Quality system: the quality assurance team audits at the suppliers' site by using a guideline prepared by the company for this purpose. It includes 11 factors which are listed in Table AI.

Once the factors are analyzed, the suppliers are assigned a cumulative grade which may be A, B1, or B2, or C depending on their audit report. However, Mekong has its own internal targets for the key suppliers and they must be granted at least B1 grade to be able to continue the business. If they fall short of this grade, the corrective action and plan for improvement for the supplier is proposed by the company.

- Products' quality and its performance: the conformance of quality as agreed on the specification is evaluated, and quality defects of material found in production are recorded for scoring. The lower number of non-conformance fetches higher score for the supplier.
- Quality improvement and responsiveness: responses to the improvement are evaluated in terms of speed of response to the quality issues or improvement opportunities provided. Quick responsiveness of the suppliers and their feedback to

Criterion	Ownership (Department who owns the process)	Frequency of evaluation
Quality system	Quality assurance	Monthly
Products quality and performance	Quality assurance	Monthly
Quality improvement and responsiveness	Quality assurance	Monthly
Technical service	R&D	Yearly
Delivery	Logistics	Monthly
Financial documentation	Accounts	Yearly
Price	Purchasing	Yearly
Management support	Purchasing	Yearly

Source: Based on the company's data (Year 2015)

Table II.
Criterion for suppliers' evaluation, their ownership and frequency of evaluation at Mekong Thai holding company

the quality improvement need is recorded regarding the number of days taken, the lower the number of days in providing feedback by the supplier, higher is the score.

- (2) Technical service: in this dimension, the speed of responses within agreed time (including fixing the problems) and quality of work, e.g., providing the information of product specification, materials safety data sheet, certificate of analysis and technical information is considered for scoring.
- (3) Delivery: this dimension is the key source of scoring. Also, the effectiveness of the delivery plan, required communication, i.e., prior advise on the delivery change if required, the delivery plan conformation within agreed timeline, and stock data accuracy are considered and counted as delivery performance.
- (4) Financial documentation: documentation of invoice and purchasing order, i.e., the accuracy of calculation for the invoice, tax invoice, promptness of billing and settling claims (credit/debit notes) are also important.
- (5) Price: there are six areas related to the price which are: price flexibility due to time elapsed or fluctuation of the price of material composition; pricing system, i.e., suppliers' cost structure (materials cost, operation cost and delivery cost); competitive pricing with comparison within the same industry with at least two suppliers; suggested cost improvement methods which is the recommendation of cheaper alternative source of materials while maintaining the same or better quality; speed of response to quotation and problem solving; and payment terms which are longer are rated higher in the score.
- (6) Management support: management support is also important and developing a long-term relationship with the supplier is a crucial part of pursuing business strategy and developing a viable value chain. Management's incessant support in forging a long-term relationship with the supplier is mandatory for an effective supply chain management.

The full scores for each dimension are allocated as per the guidelines given in Table III. The company assigns these scores religiously, and it is seen as an important and fundamental exercise of the suppliers' management program at the Mekong. The raw materials suppliers are divided into two groups, i.e., specialty and non-specialty. The specialty category of the suppliers is those who can provide technology or know how to the company which is evaluated on the technical support. While the non-specialty suppliers are those suppliers who cannot contribute any technology to the company because their raw materials are simple such as salt or soya bean oil. They are evaluated on the dimension of the technical support area and fetch fewer scores than specialty group.

After scoring each dimension for the key suppliers' evaluation, suppliers are classified into three levels as preferred suppliers, acceptable suppliers and unacceptable suppliers. The preferred supplier is the supplier having a score of 140 or more for specialty category

Table III.
Scoring criteria and
full score for specialty
and non-specialty
product suppliers

Section	Specialty	Non-Specialty
Quality	55	55
Delivery	30	30
Technical support	20	—
Financial documentation	5	5
Price	50	50
Management support	15	15
Total	175	155

Source: Based on the company's data (Year 2015)

and 124 or more for the non-specialty category. The preferred supplier gets the priority while putting new purchasing order for the new raw materials. The acceptable supplier is the supplier having the score in the range of 106–139 for specialty category and 94–123 for the non-specialty category. The Mekong finds that they can do business with this kind of suppliers but they are not seen on the priority list to be considered. The last category of the supplier is unacceptable suppliers who have a score of 105 or less for the specialty category and 93 or less for the non-specialty category. The company stops doing business with this kind of suppliers. However, in some cases where there are some constraints which prevent a sudden termination, an improvement plan is proposed to be undertaken by the supplier at the earliest (Table IV).

Step 5: supplier appraisal and the feedback process. After the scores are obtained, they are communicated to each key supplier in an appraisal session which is held between February and March of the following year. The appraisal sessions let the company's cross-functional team talk directly with the suppliers' top management team on their performance on the criterion, and discuss their strengths and weaknesses to improve the gap if any. It also delineates the necessary support system which suppliers would like to get from the company. There are five steps in the appraisal process. First, supplier performance is recorded by the cross-functional team on five dimensions (Quality: QA, Delivery: Logistics, Technical support: R&D, Financial documentation: Accounts and Price: Purchasing) during the appraisal period, January–December. Second, each department completes the suppliers' assessment by the end of January of the following year. After that, scores are reviewed and discussed. The evidence is documented properly to maintain the validity of the score. In the fourth step, scores are aggregated, and suppliers' categories are determined as preferred, acceptable, and unacceptable depending on their scores. Finally, the decisions are made during the month of February and March by the Mekong Thai Holding Company regarding retaining the suppliers' status based on their achieved scores.

As presented in Figure 2, the total score of key suppliers is given. In the year 2015, it was found that out of 18 key suppliers, 13 were chosen as "preferred suppliers" and five as "acceptable suppliers" based on their scores.

The current study clearly demonstrates that evaluating suppliers and supporting their continuous improvement is critical to the supply chain management. At the Mekong, the key suppliers' management program was implemented in the year 2015. The case study collected and compared the past six years' data for the period 2010–2015. It can be observed that the performance of the suppliers improved in general after adopting the suppliers' management program. As evident (see Figure 3), all key suppliers' performance was improved, and an increasing % OTQIF trend was observed.

It can be observed from Figure 3 that supply chain irregularity was rectified to a great extent after adopting the supplier's management program at Mekong Thai Holding. It can be seen that the incomplete delivery data were reduced in the year 2015 substantially as compared to the years 2010–2014. Figure 4 shows the decreasing numbers of incomplete deliveries in the year 2015 compared to the years 2010–2014, and it reinforces the efficacy of the suppliers' management program.

Category of supplier	Score for specialty product suppliers	Score for non-specialty product suppliers
Preferred	140 or More	124 or More
Acceptable	106–139	94–123
Non-Preferred	105 or Less	93 or Less

Source: Based on the company's data (Year 2015)

Table IV.
Categorization of
suppliers based
on their score

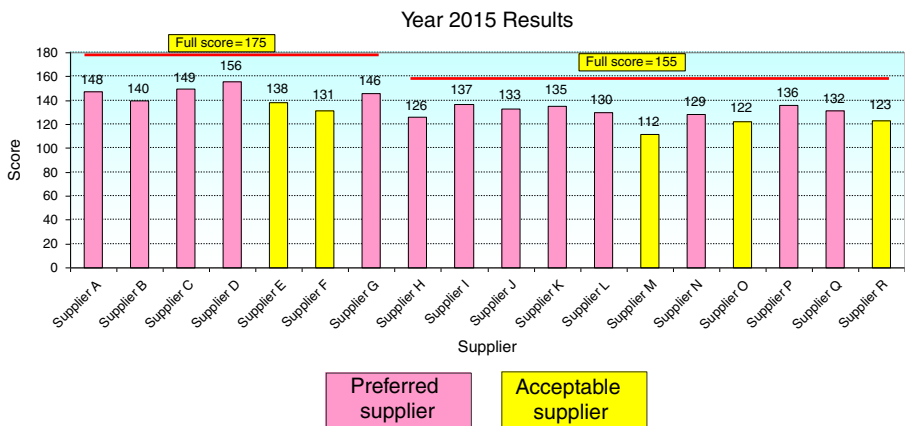


Figure 2.
Suppliers' score
(year 2015)

Source: Based on company's data year 2015

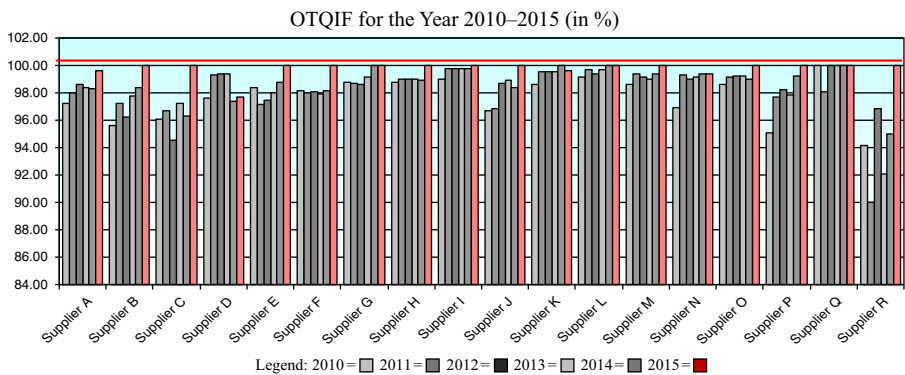


Figure 3.
Comparison of OTQIF
of suppliers during
the years 2010–2015
(in %) before and after
the supplier selection
and management
program implemented
at Mekong

Source: Based on company's data year 2015

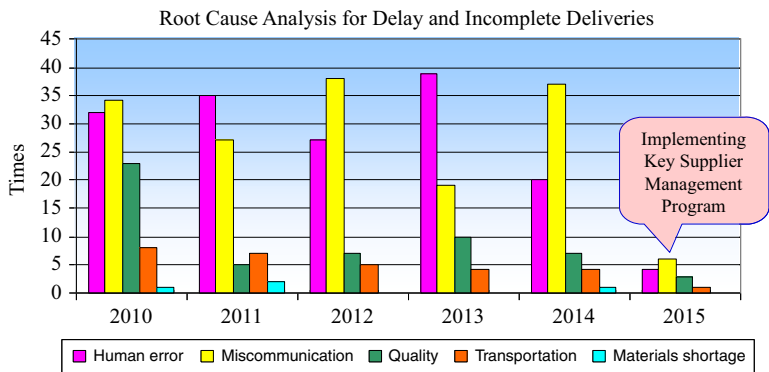


Figure 4.
Comparison of
missing lot causes
before and after the
implementation of
supplier management
program at Mekong
Thai Holding
Company (years
2010–2015)

Source: Based on company's data year 2015

As discussed in the earlier section of the paper (from steps one to five), suppliers' involvement helps in improving the supplier's management program. It was validated by this research that when suppliers perceive the benefits of any suppliers' evaluation program they dedicate their efforts and resources effectively to eliminate the problems that are revealed in their scores. After cause analysis of the missing lot, the company found an obvious reduction in the total numbers of such cases. The supplier appraisal program was found to be very fruitful, and the company got inputs from the suppliers' feedback. Involving and sharing appraisal data resulted in bridging the gap, and improved the quality of supplies. It helped in reinforcing a climate of trust between the suppliers and the firm at Mekong. It can be said that five main reasons found to be playing a role of catalyst in improving the quality of relationship with the suppliers at Mekong were as follows:

- (1) First, top management support and encouragement facilitated in creating a climate of trust and ownership between various stakeholders. The cross-functional team played an effective role in pursuing the quality aspect of supply management religiously. Top management concern and commitment provided the requisite push to deal with the roadblocks and problems which are part of any team playing exercise.
- (2) Second, involving suppliers in managing supply chain was seen as an important function. The selection and appraisal process of suppliers at the Mekong Thai Holding company was instrumental in creating an open and transparent evaluation system involving all stakeholders. Having an honest and transparent appraisal system helped the suppliers in ascertaining their strength and weaknesses from the company perspectives.
- (3) Having a regular feedback process with suppliers helped them in improving their performance regarding timely delivery with the requisite quality with no short supply or short shipping. Communicating the supplier their category as preferred or acceptable was seen as a reward by the suppliers. Those who were categorized as preferred suppliers saw this as a token of their accomplishment and reward, and they went beyond in forging long-term relationship by contributing the firm's value chain stream positively.
- (4) Fourth, having the suppliers' appraisal system and metrics in place helped the Mekong Thai Holding company in integrating its operation. Continuous interaction of quality, R&D, logistics, accounts and purchasing personnel helped in improving the quality and cost aspects of the operation. Resultantly, it improved the overall performance management system of the firm.
- (5) Finally, the continuous interaction and exchange of ideas with the suppliers helped in product development and new ideas generation. It also helped in cutting the cost and improving the quality of the products. Sharing best practices helped both parties in enriching their experiences and bringing effectiveness in their operation system.

Conclusion

Adopting an effective suppliers' management program is critical for selecting the best suppliers and improving their performance over a period of time. Developing metrics for suppliers' performance is the key to achieving better results as evident in this case. Measurement tells the suppliers what is valued by the organization and what is not. Without measuring their performance, it is impossible to tell that performance of a system is getting better or worse. Failure to accurately measure and evaluate suppliers' performance increases an organization's costs, and damages its product quality and hinders in acquiring competitiveness. Supplier performance information is vital for making optimal supply chain management decisions.

Suppliers' selection and management help an organization to reinforce its competitive advantage from suppliers' point of view. Closer communication and setting a measurable common target for both parties results in the performance improvement in the supply chain management. Furthermore, an evaluation system which is transparent and open gets communicated to suppliers, and helps them to set direction and to work on an action plan for continual improvement toward a sustainable target. However, the factor that influences success in a suppliers' management program is institutionalizing a win-win culture promoting an open, trustworthy, reliable and ethical partnership. Since a supplier measurement system impacts the entire organization, success is more likely if all stakeholders are involved. Having a cross-functional team which is well groomed and have gone through the various stages of team building process is necessary. Much depends on the success of the effective cross-functional team consisting of various functions. This fosters the perception that the system is well integrated and suppliers' selection and management is not only the responsibility of the purchasing department, rather this responsibility is owned and shared by all concerned departments comprising QA, logistics, R&D and accounts as found in this case. When organizations decide to develop their own systems, the team must work with system developers to ensure the desired outcome.

An organization needs to identify its business priorities before a strategic decision is made regarding the adoption of a suppliers' selection and management program. Implementing a supplier management program is a critical step in enhancing an organization's competitiveness and it demands the commitment and urgency of the task in that regard. The support and commitment of the top management are one of the key ingredients for the success of any suppliers' management program. Involving all stakeholders is important and developing a cross-functional team is a requisite. Organizations must encourage a culture of teamwork and collaboration for succeeding in any such endeavor. A supplier management program is likely to be used to manage and control the supplier's contract performance and for the sourcing selection as well for the future contracts. Organizations need to continuously monitor, measure and analyze suppliers' appraisal program and share its results with their suppliers. Moreover, a preview and testing phase before the system implementation provides the opportunity to discover any gap in the existing system. It familiarizes users with the system and enhances the sense of ownership among the internal customers. During this phase, purchasing department must communicate effectively both with the internal customers and with the suppliers to develop buy-in and removing any perception blockade.

The case study helps in explaining the selection and evaluation of the suppliers' performance as an integral part of the business relationship. Organizations are supposed to be proactive in their suppliers' management most of the time. In the absence of a rigorous suppliers' management system, measurements done are also not up to the requisite standard and does not help much in improving the relationship with the suppliers. The continuous measurement of the supplier, its documentation and sharing it with the suppliers is an important task in this exercise. It has been found that many a time, the performance of the suppliers is not monitored continuously and the problems are not foreseen and identified at the outset. The goal of the suppliers' selection and management program is to find and develop sources that can meet current and forecasted operational and strategic needs. Evaluating suppliers is a costly exercise as is switching the suppliers. Typically, organizations preferred to avoid excessive supplier turnover without becoming complacent. Maintaining a balance between fostering competition and fostering co-operation is a challenge for organizations. The long-term partnership which is comprised of trust is needed to ensure supply reliability for the organization.

Managerial implications

On the practical side, the study provides an avenue to explore the strategic decision in supplier selection strategy and supplier management strategy concerning organization requirements because different types of the organization require different drivers in supplier selection and

supplier management. There is no single formula that applies to every situational context. An effective supplier management system provides a better and more dependable delivery service to the manufacturer. When production occurs at different locations, manufacturers can determine quickly the most economical source for a particular facility. Manufacturers expand into international markets and source these markets from multinational production facilities so that they can confront the cost variables that make it imperative to employ an effective supply management program to achieve an efficient operation.

The present study shows that after the implementation of a supplier's management program, the suppliers' performance becomes much better. The results imply that after selecting the right suppliers, it is necessary to share with them the right organizational requirements in terms of quality, cost, delivery and technical support. The dynamics of the suppliers' management program and its outcome must be shared with the suppliers in the appropriate time frame. Establishing a long-term relationship with the supplier is a prerequisite for the organization, and suppliers have a direct impact on the competitiveness of the firm. Institutionalizing an effective supplier management program contributes effectively toward helping firms in achieving their long-term competitive advantage.

Limitations and future research

There are several limitations of this study. First, from the methodological standpoint the study has been conducted using a single case company which makes the study highly contextualized. Second, the data for the study have been gathered within one specific type of industry, e.g., FMCG. This limits the applicability of the findings to other types industries. The nature of demand for these products is quite different from other products. Therefore, the findings of this research may not be generalizable to other industries. Third, this case study has been developed within one single developing economy in Southeast Asia. From the consumer behavioral perspective, one of the challenging aspects of the FMCG industry is that it is highly influenced by culture. That means the buying behavior is different from other parts of the world. Hence, replicability of the findings of this study might be restricted to the region where the study has been carried out.

Despite these limitations, this study provides some deeper insights to improve efficiency across supply chain. Based on the current study to overcome the above limitations, after suppliers acknowledged the opportunities for improvement for the key suppliers' management scores, the future research should focus on how to initiate that improvement in the specific area, e.g., in delivery performance by using IT tools, quality performance by having early suppliers' involvement; pricing – by helping suppliers to reduce their cost by requesting suppliers to share their cost structure and help them to reduce the overall cost. Another avenue for future research should be to measure these performance dimensions in a longitudinal perspective and investigate their variations over time, i.e., seasonal effect. Finally, the future research should apply and investigate these suppliers' management dynamics across other industries and empirically validate them.

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Appendix

Quality system factors	Criterion for evaluation
General requirement	Quality policy, quality manual, internal audit, and legal compliance
Factory design and the layout	Building and its facilities, manufacturing area, laboratories (internal or external), lighting and climate control, equipment design and maintenance
Raw and packaging materials	Selection and management of suppliers, raw materials and packaging specification, receipt, and storage of materials
Processing and packaging	Process control, work procedure and training, calibration, control of foreign bodies/contaminations, non-conforming product and trade return, rework, lot coding and traceability
Finished products	Agreed on the specification, document control, conformity control, product release procedure, complaint management and feedback
Storage, transportation and distribution	Transportation and transport contract, storage facilities
Hygiene, cleaning and disinfection, pest control	Cleaning and disinfection procedures, housekeeping and cleanliness, waste management, and pest and insect control
Personnel and their training	Personal hygiene facilities, health screening, personal training, and developing the competence
Environmental care	Compliance with legislation, environmental management system, and continuous improvement of environmental impact
Health and safety at work	Compliance with legislation, health and safety at work management system, safe and healthy working conditions for all employees
Code of business principle	No use of any form of forced, compulsory or child labor, compliance with the law, business integrity (no bribery)

Table AI.
Quality system
evaluation criteria

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